환경과학개론 (EVSE101)

Part 2-7. 환경-에너지 넥서스 기술: 탄소 중립과 지속가능성

2025년 2학기 후반부

1 분반: 월 수 11:00 – 12:15
2 분반: 월 수 12:30 – 13:45
청암학술정보관 멀티미디어 교육실 501호

물-환경-에너지 넥서스
(Water-Environment-Energy Nexus)

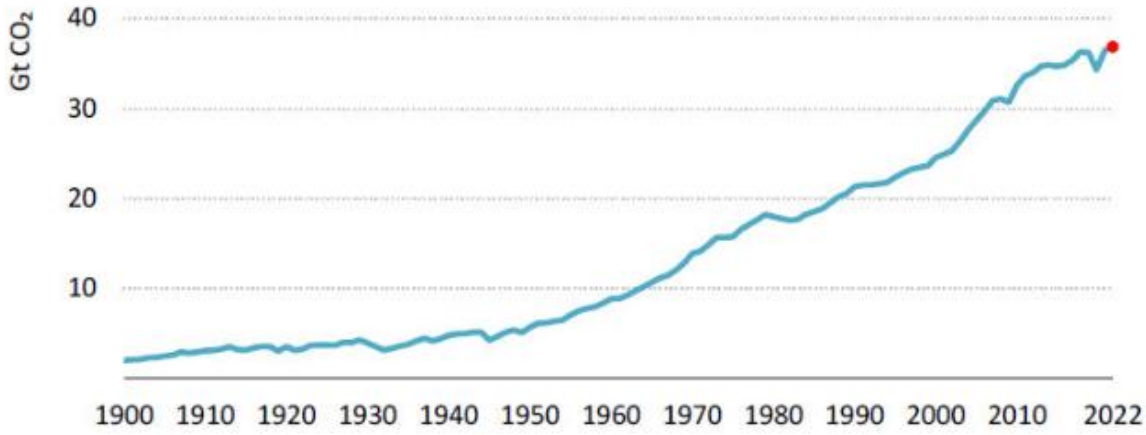


지속가능성(sustainability)?



Rita A. Bajura is director of the National Energy Technology Laboratory. This article is based on a presentation given at the Technical Session of the NAE Annual Meeting in October 2001.

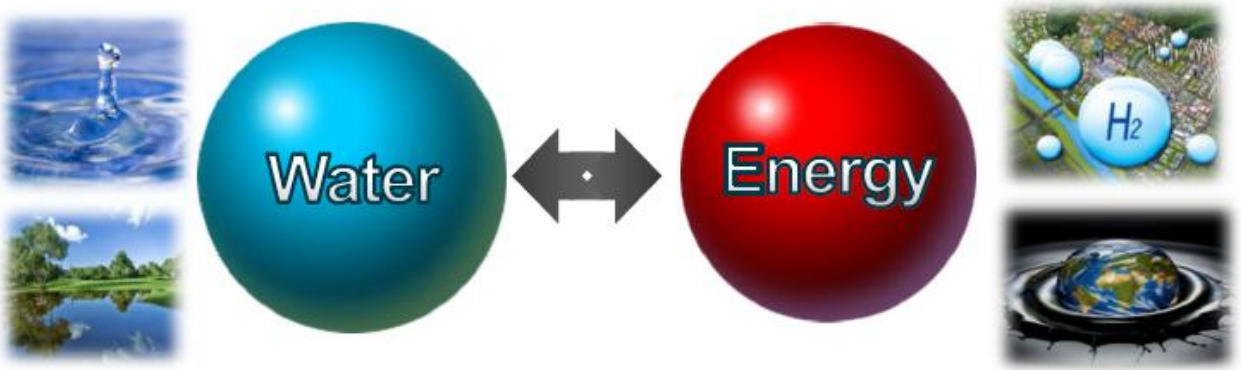
The word "sustainable" means different things to different people. To some, it means integration of the social, economic, and environmental domains. To some it means energy that lasts forever. And to some it means energy with no environmental cost for its production and use. Energy from every source has environmental or cost consequences. The challenge for us as a society is to agree on a practical definition of sustainability and then to develop a road map to achieve it. The road map should include public policies, incentives, and research and development (R&D) agendas.

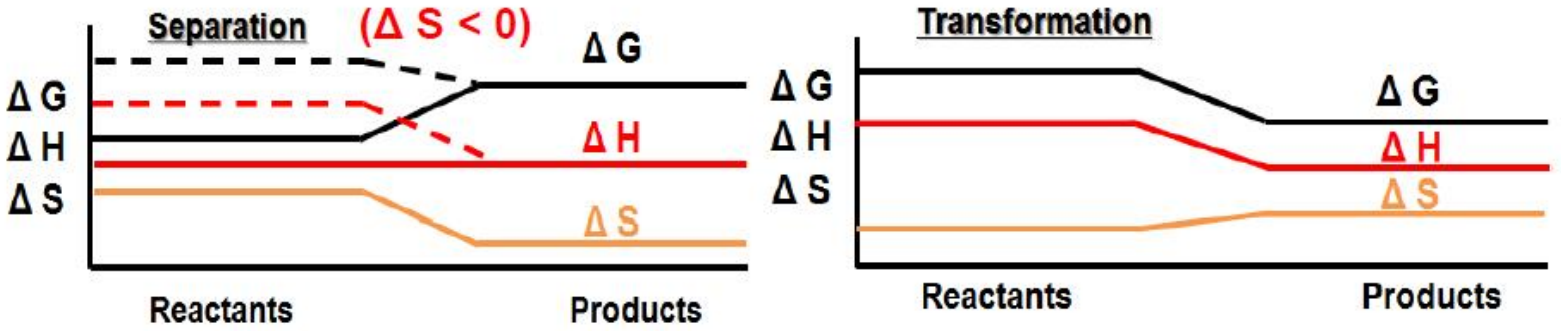


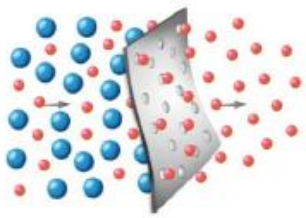
1990~2022년 에너지 연소와 산업 공정에서 배출되는 전세계 이산화탄소 배출량

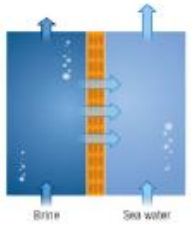
출처: 국제에너지기구 2022년 이산화탄소 배출량 보고서

물-환경-에너지 넥서스 (Water-Environment-Energy Nexus)



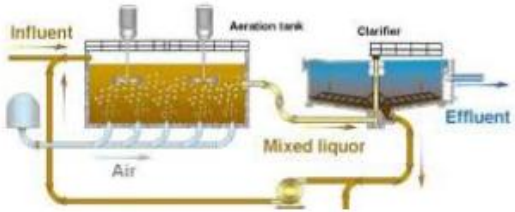






$$\Delta G = \Delta H - T\Delta S$$

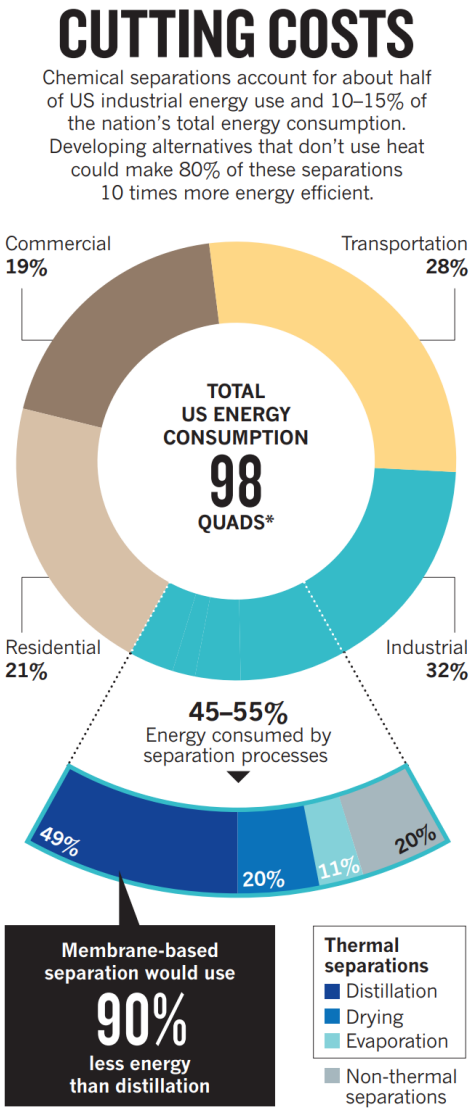
$$H = U + pV$$



분리 기술

❑ 전세계 에너지 소비 중 분리공정이 차지하는 에너지 사용 비율: 15%

❑ 화학 및 정유 플랜트, 제약회사 등 운용 비용의 70%가 분리공정에 사용



출처: Science 저널

분리 기술 – 막분리

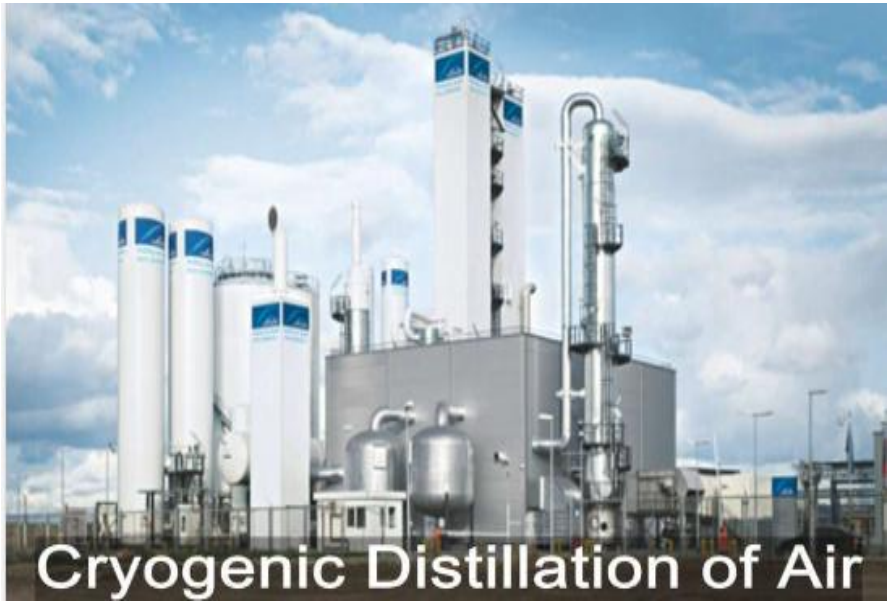
CO_2 제거

Amine Scrubbing tower

분리막 모듈



흡수 vs 분리막



상 전이 - 열공정 (Phase change)



증류

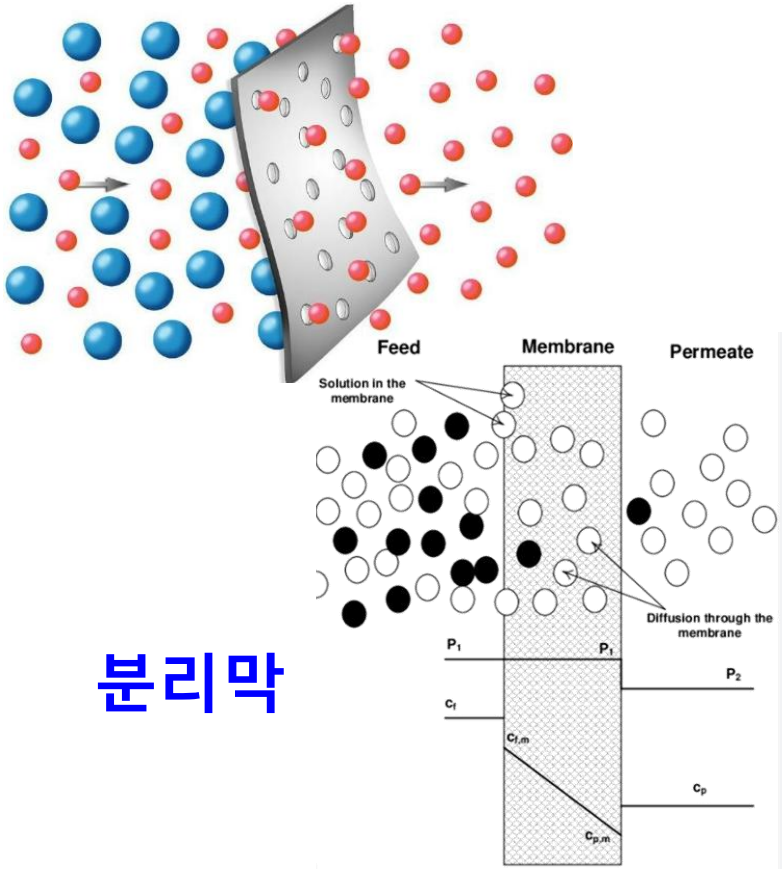


흡수착

에너지 소비 큼

VS

소재 내 물질 이동

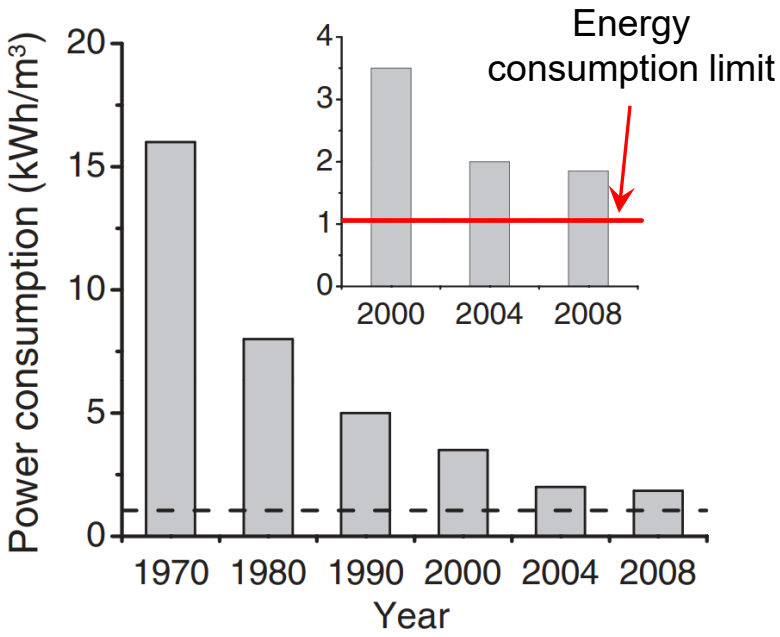


분리막

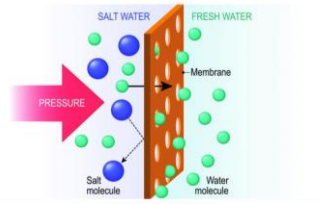
에너지 소비 작음

역삼투 해수 담수화





역삼투 분리막



Reverse-osmosis (RO)

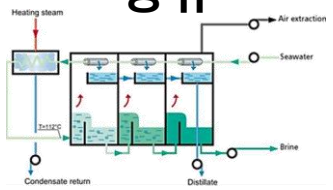


~0.6 M Ton

per year

(3 ~ 6 kWh/m³)

증류



Multi-stage flash (MSF)



5천만 인구에게 물 공급 시

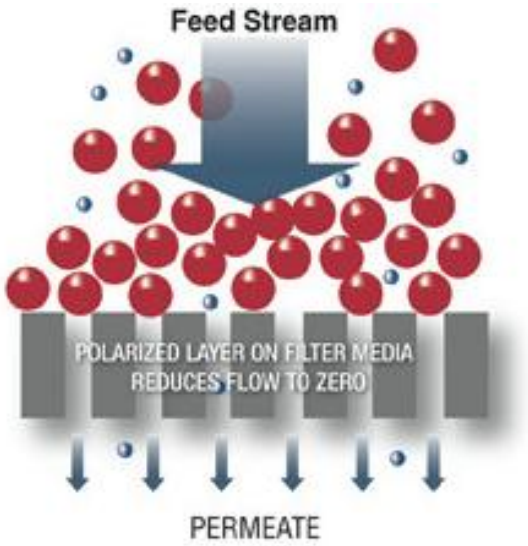
>4 M Ton

per year

(10 ~ 30 kWh/m³)

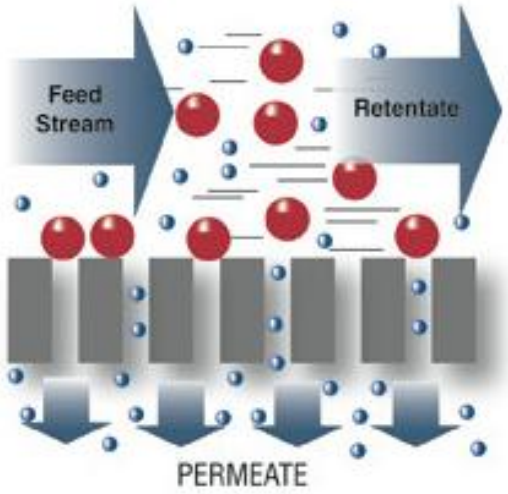
분리막 투과 (여과) 방식

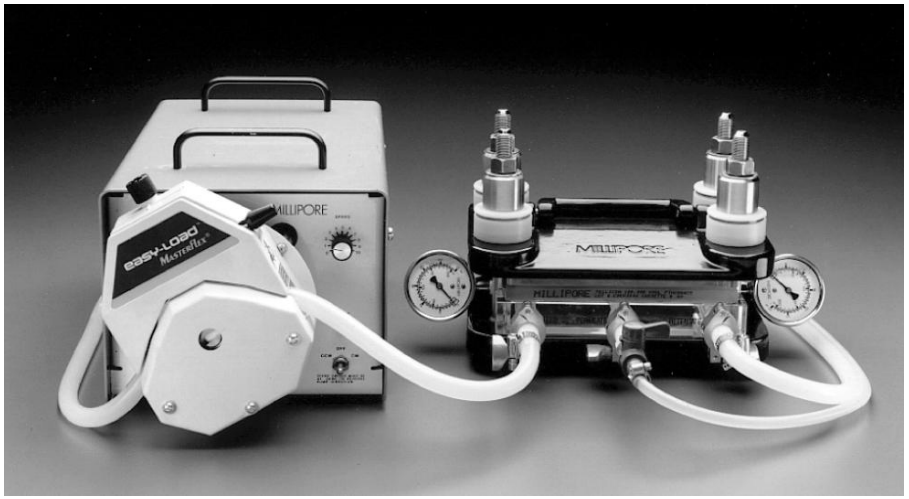
데드 엔드 (Dead-end)





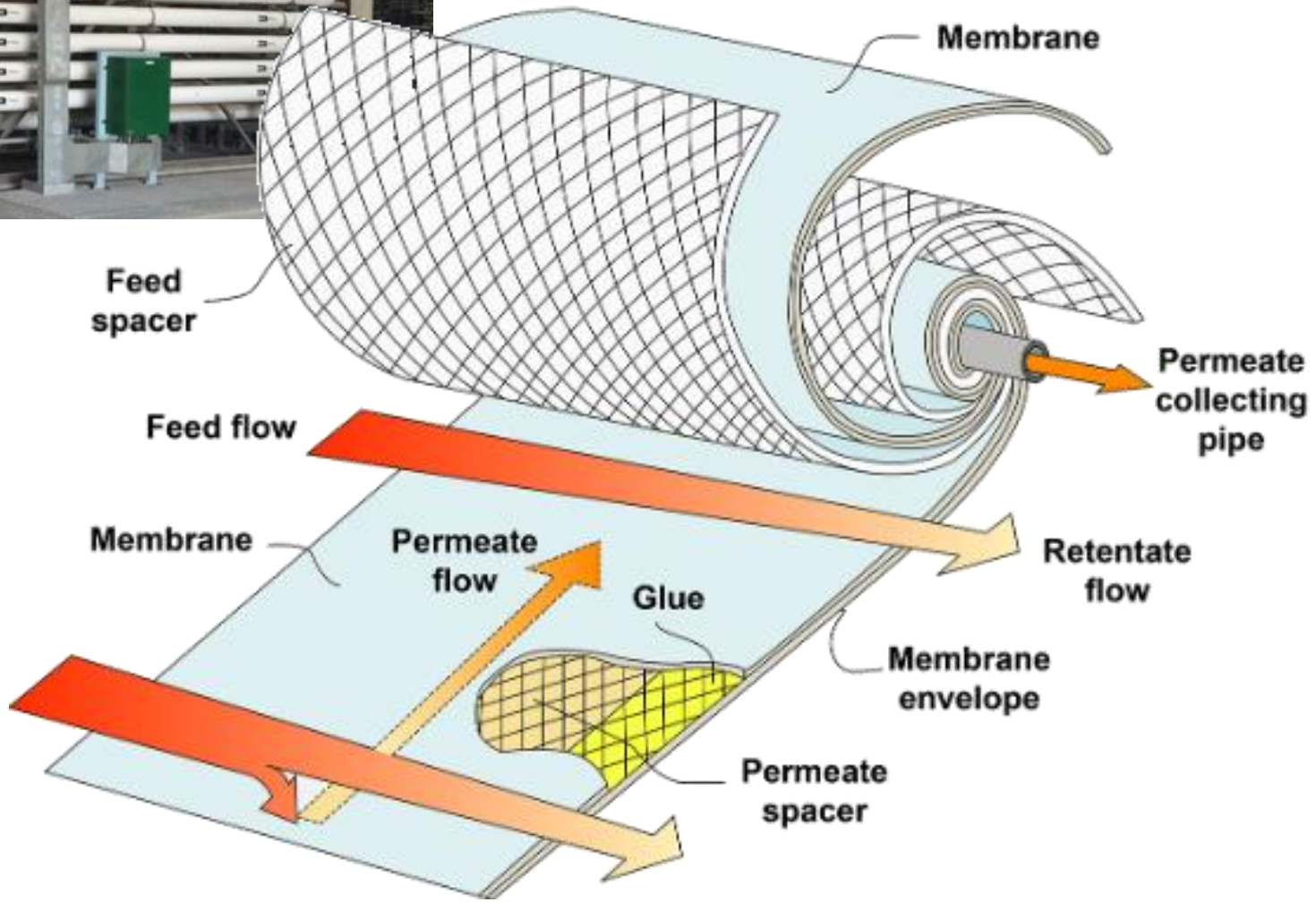
크로스 플로우 (Cross-flow)



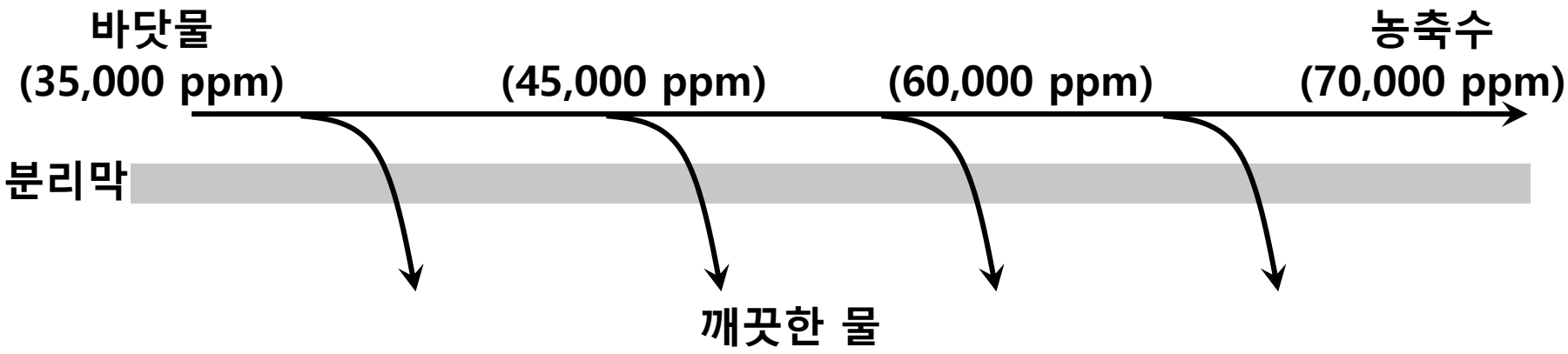


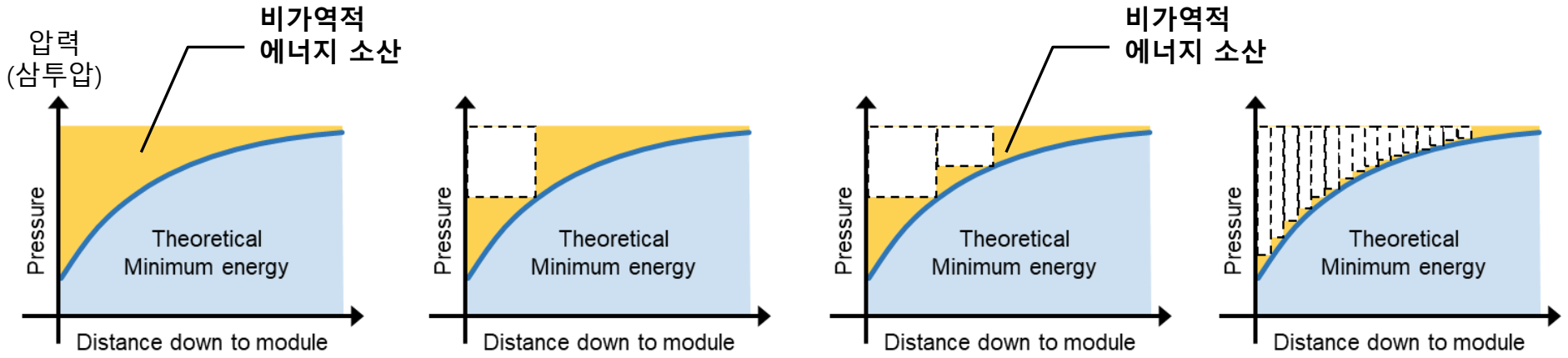
분리막 모듈

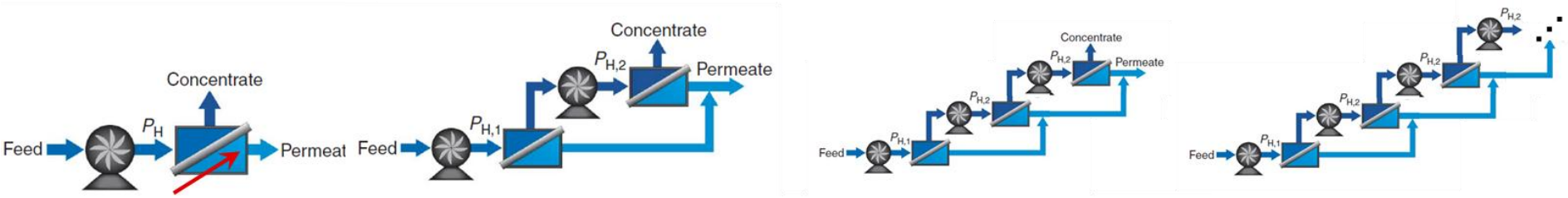



Membrane
Feed spacer
Permeate collecting pipe
Feed flow
Membrane
Permeate flow
Retentate flow
Glue
Membrane envelope
Permeate spacer

역삼투 해수담수화 – 공정 최적화?







Joule

CellPress

COMMENTARY

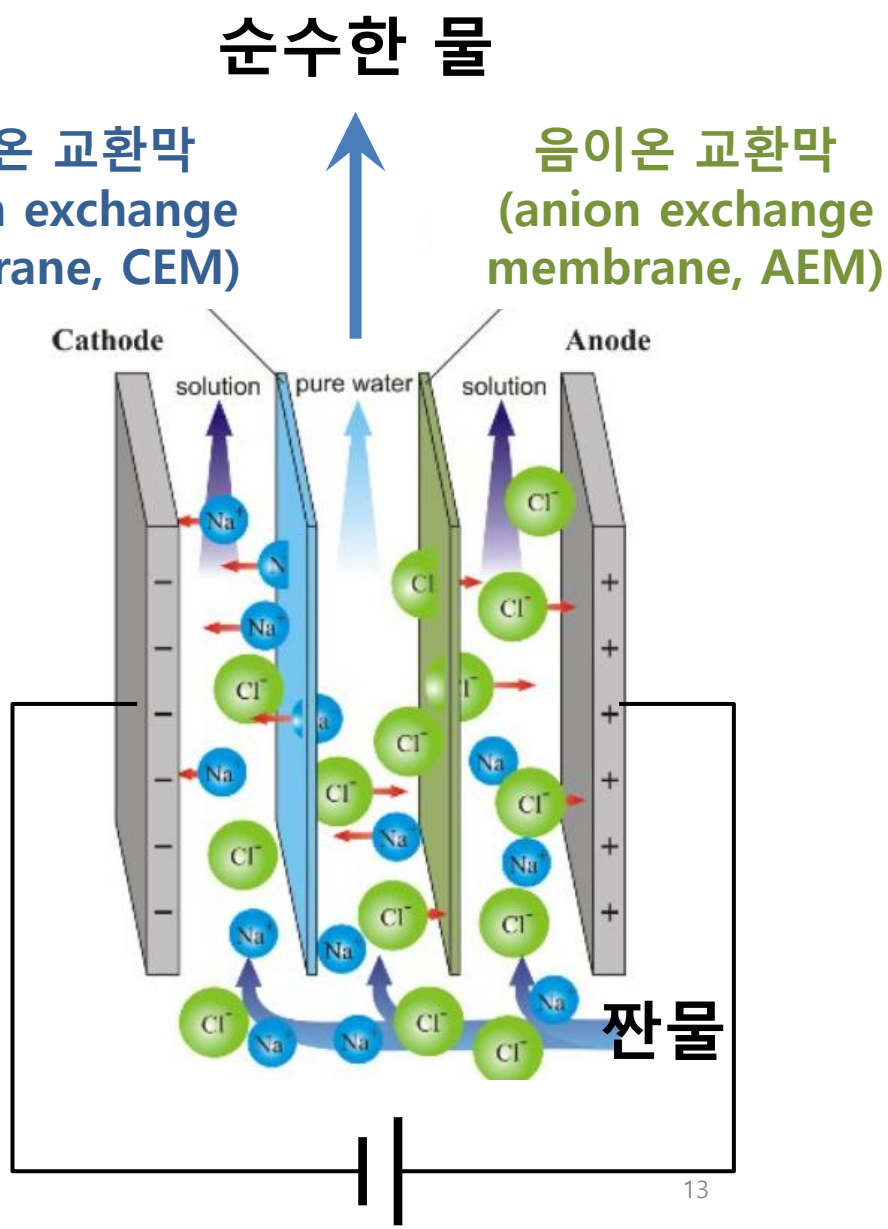
Avoiding the Hype in Developing Commercially Viable Desalination Technologies



Jeffrey R. McCutcheon[1,*]

1168 Joule 3, 1163–1171, May 15, 2019 © 2019 Elsevier Inc.

전기 투석 (Electrodialysis, ED)

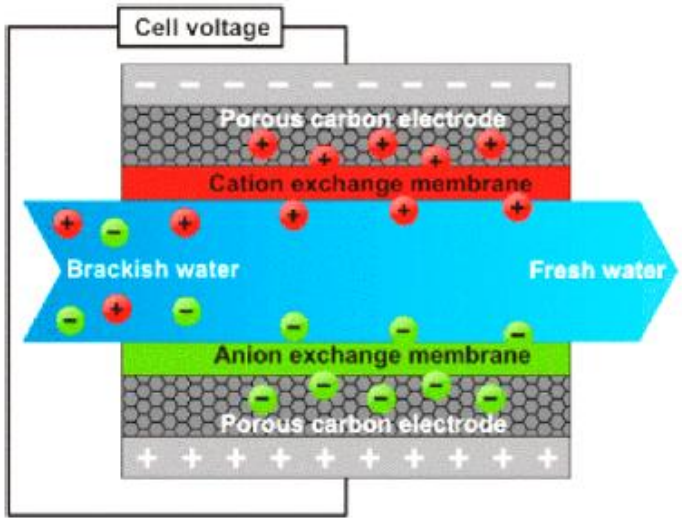




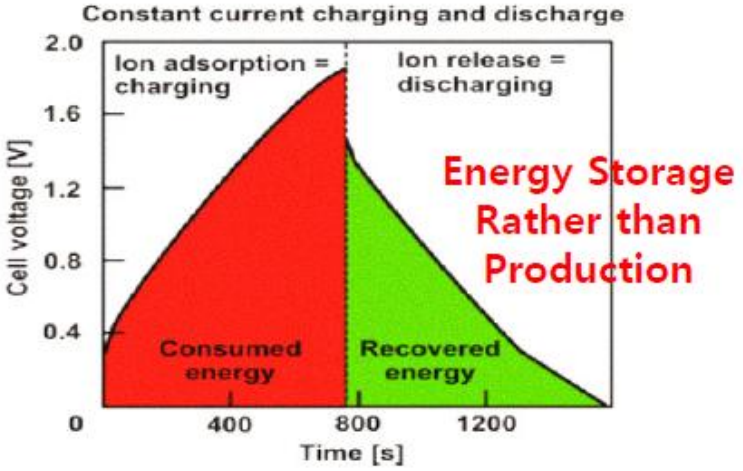
전기투석조(Electrodialysis Stack)<출처 : (주)이노메디텍>



축전식 탈염 (Capacitive Deionization, CDI)



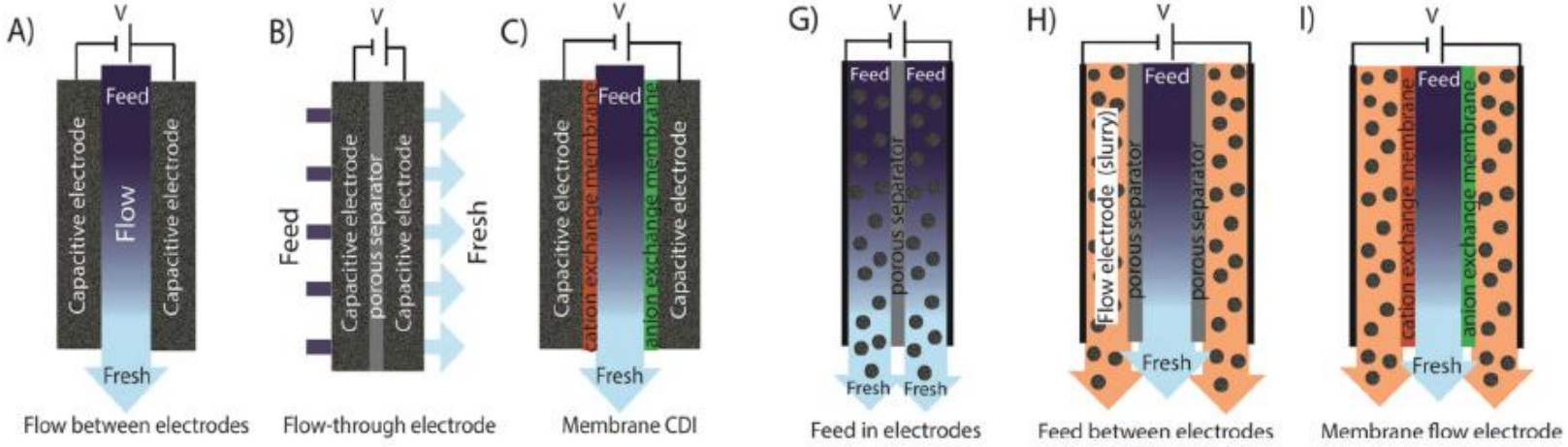
Długołęcki and van der Wal (2013) *ES&T*, **47**, 4904



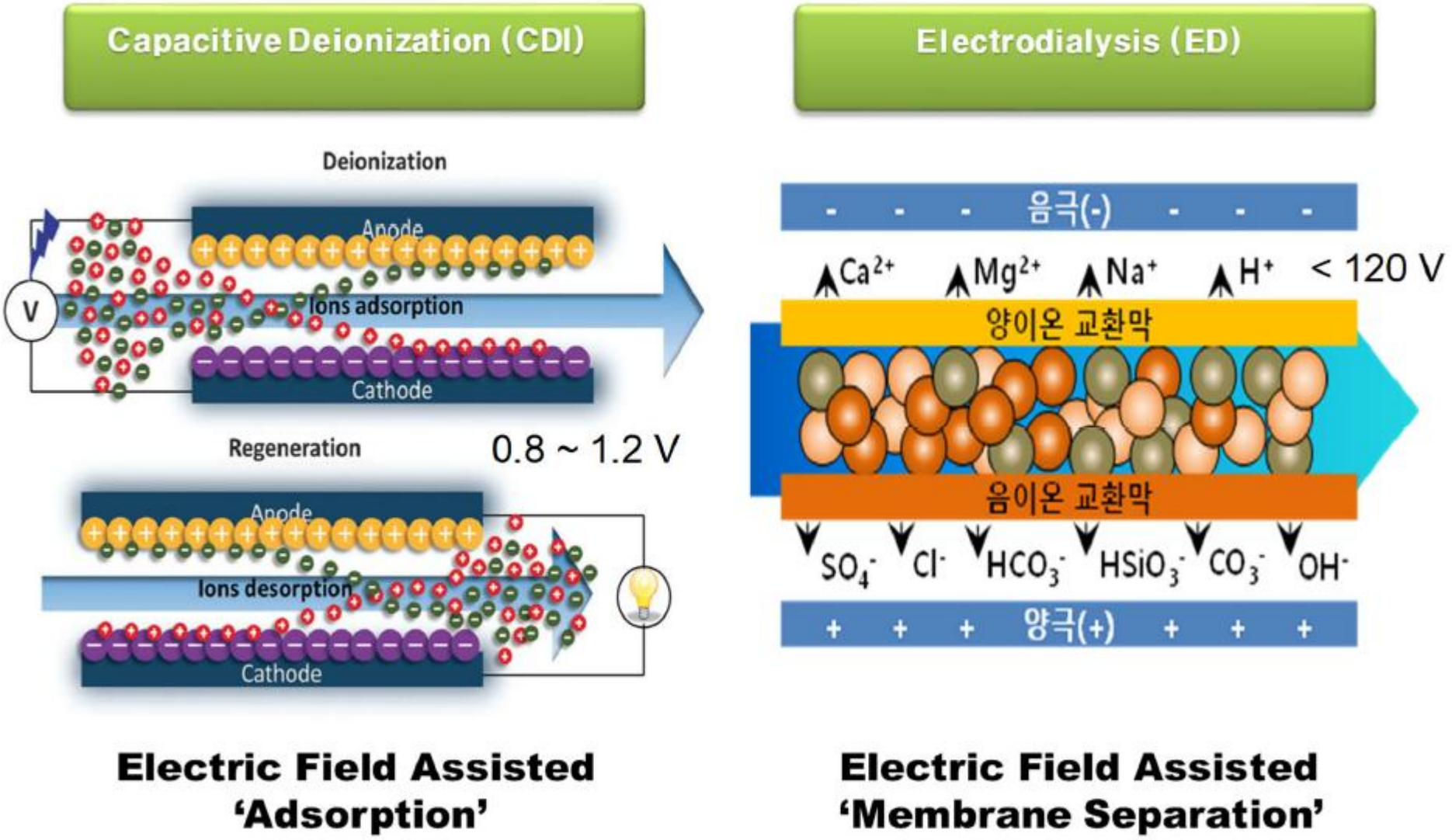
Suss *et al.* (2015) *Energy Environ. Sci.*, **8**, 2296

Static electrode architectures

Flow electrode architectures



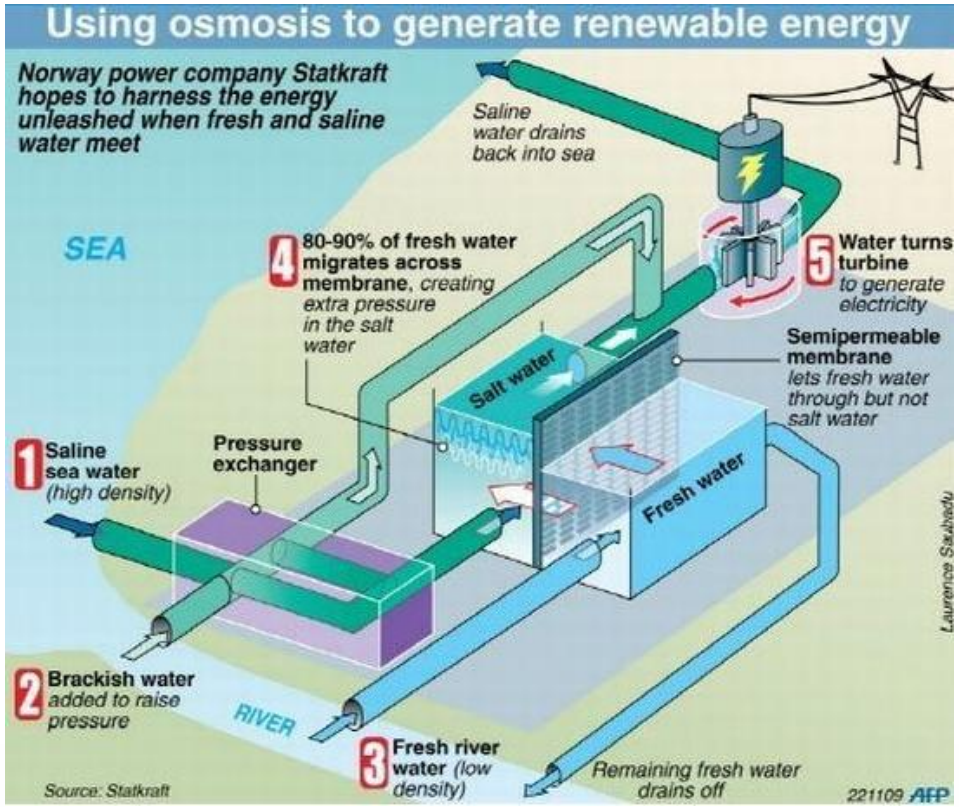
축전식 탈염 (CDI) vs. 전기투석 (ED)



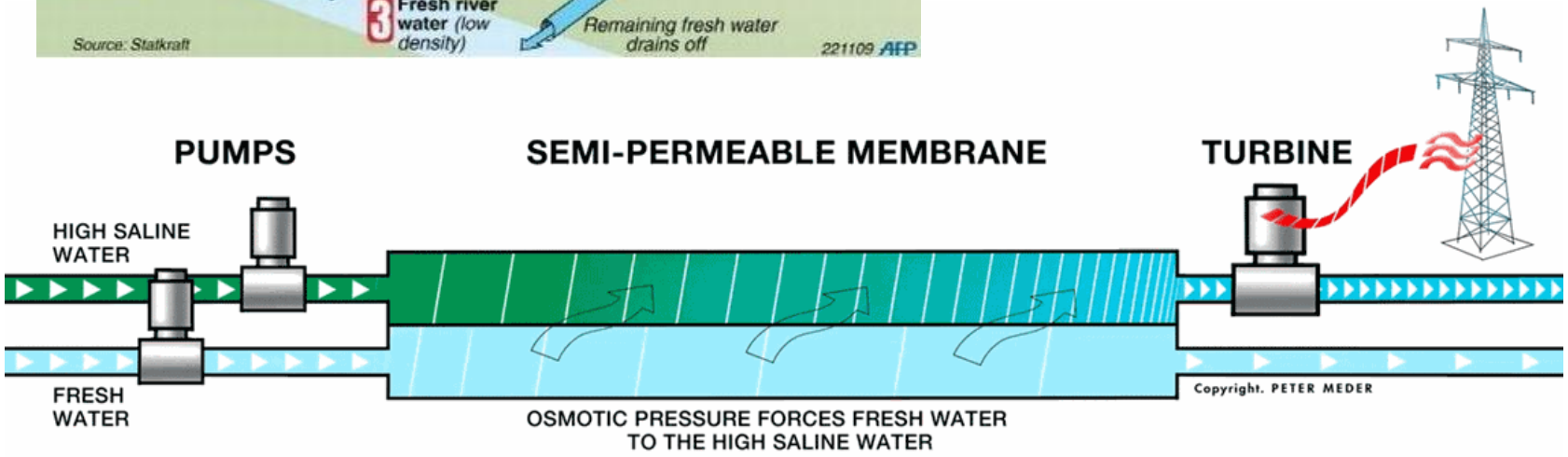
Electric Field Assisted 'Adsorption'

Electric Field Assisted 'Membrane Separation'

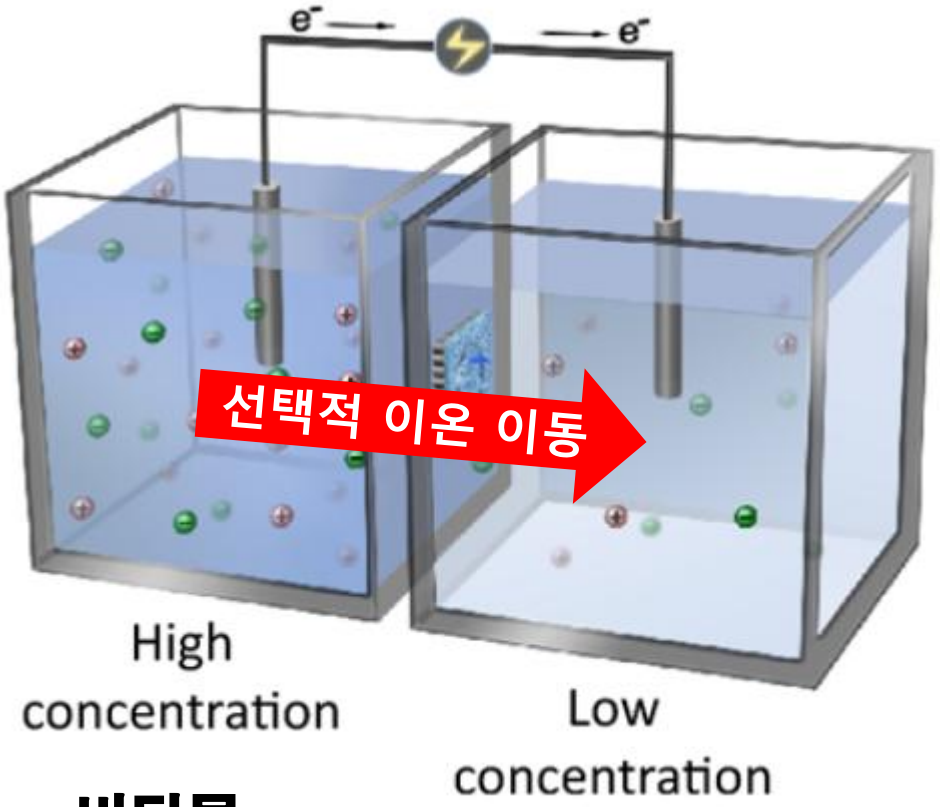
압력지연삼투발전 – 블루 에너지 하베스팅







염도차 발전 – 블루 에너지 하베스팅



바닷물

강물



염도차 발전 기술의 핵심

- 이온 선택성
- 이온 투과도

현재 구현 된 에너지 생산 밀도

나노스케일 : ~2000 W/m^2

매크로 스케일 : ~5 W/m^2

미생물 연료 전지 (microbial fuel cells, MFC)

https://www.youtube.com/watch?v=GTVoEVM4Z8s



